KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Mid Semester Examination, March, 2021
Third Year

TE 385: Signals and Systems ***TIME: 1 hour***

Program of study: TE INDEX NO:

Attempt all questions

SECTION A (10 Marks)

Fill in the blanks

1 Signals that have finite signal energy are referred to as Energy signal

2 The homogeneous solution to a differential equation is often referred to as natural response

3 A shift of a signal to the left is a Advance

4 When a periodic signal is shifted in time, the magnitude of its Fourier Series coefficients remain the same

5 Discrete-time signal may be obtained by Sampling continuous-time signal

6 The convolution of a signal with an impulse is equal to Original signal

7 $t\delta(t) =$ 0, for all values of t.

8 If $x(t)$ is replaced by $x(at)$, then $x(at)$ is linearly compressed version of $x(t)$ if $|a| > 1$

9 Any discrete-time signal can be expressed as a sum of time-shifted impulses

10 For an odd signal $x(-t) =$ $-x(t)$ for all t.

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TE 385: Signals and Systems

INDEX NO: 3569618

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SECTION A (10 Marks)

Fill in the blanks

7/10

1 Signals that have finite signal energy are referred to as ....Energy Signals....

2 The homogeneous solution to a differential equation is often referred to as

.....X.....

3 A shift of a signal to the left is a ....~~delay~~ Advance....

4 When a periodic signal is shifted in time, the magnitude of its Fourier Series coefficients

remain the same

5 Discrete-time signal may be obtained by ....Sampling.... continuous-time signal

6 The convolution of a signal with an impulse is equal to ....weighted sum ~~of~~ X individual Impulses

7 $t\delta(t) =$ ....0...., for all values of t.

8 If $x(t)$ is replaced by $x(at)$, then $x(at)$ is ....linearly ~~stretched~~ compressed ~~Compressed~~ ~~Stretched~~.... version of $x(t)$ if $|a| > 1$

9 Any discrete-time signal can be expressed as ....X.... of time-shifted impulses

10 For an odd signal $x(-t) =$ ....$-x(t)$.... for all t.

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7 Consider

SECTION B (50 Marks)

1 What is a periodic signal? Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x(t) = 5\cos(40\pi t + \frac{\pi}{6})$

(b) $x[n] = 3\cos(6\pi n + \frac{\pi}{6})$

2 Consider the system $\mathbf{y(n) = Cos\, x[n]}$, state whether the system is

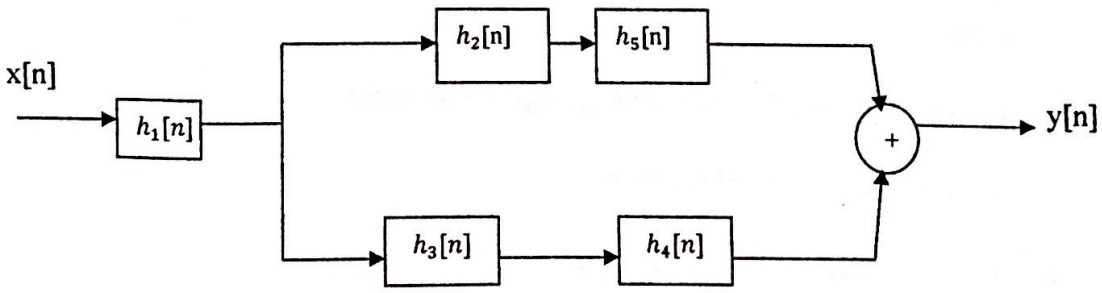
(a) Memoryless
(b) Linear
(c) Time invariant
(d) Causal

Justify your answers

3 Compute the convolution of x(t) = u(t) and h(t) = u(t)

4 Compute y[n] = x[n]*h[n], when x[n] = {3, 1, 3, 1} and h[n] = x[n].

5 Consider the interconnection of LTI systems as shown below. Express the overall impulse response in terms of $h_1[n], h_2[n], h_3[n]$, $h_4[n]$ and $h_5[n]$



6 Convert the function $g(t) = (1+j)e^{j4\pi t} + (1-j)e^{-j4\pi t}$ to an equivalent form in which j does not appear.

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7 Consider the signal $x[n] = \{0, 1, 1, \underset{\uparrow}{2}, 2, 1, 1\}$

Sketch and label carefully the following: (a) $x[n+1]$ b) $x[-n]$ c) $x[n]u[-n]$

d) $x[n-1]\delta[n-2]$

K. A. P. Agyekum

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